

Computational Linear Algebra (MAT 2135): FISAC Assignment

B.Tech. Mathematics & Computing

Max. Marks: 10

Note: Use A4 white sheets to write answers to questions. Write your Roll number and Name on each sheet. Scan and upload it as a pdf file before the last date and time. Only submission made in LMS Platform **on or before 06 November 2025; 7 PM** will be considered for evaluation.

Q1. Consider the following tensor. Take k = Digital root of your Roll Number.

$$X = \left\{ \begin{bmatrix} 3 & 4 & ? & 1 & k \\ 5 & 1 & ? & k & 2 \\ ? & k & 2 & 1 & 5 \end{bmatrix}, \begin{bmatrix} 3 & 2 & ? & 1 & k \\ 2 & 1 & k & 4 & 5 \\ ? & k & 2 & 1 & 5 \end{bmatrix}, \begin{bmatrix} 3 & 2 & ? & 1 & k \\ 3 & 1 & 5 & k & ? \\ ? & k & 2 & 1 & 5 \end{bmatrix} \right\}$$

Let $Y = X(:, 2, :)$ be the matrix of the ratings given for three different products by five different users. In Y , Rows 1 to 3 correspond to Product A, Product B and Product C respectively; and Columns 1 to 5 correspond to User 1, User 2, User 3, User 4 and User 5 respectively. Find the

- (a) user-based prediction for the rating for Product C by User 5 based on cosine similarity.
- (b) item-based prediction for the rating for Product A by User 3 based on cosine similarity.

(2 Marks)

Q2. Take a matrix Y formed in Q1 with ratings that you have found rounded to the nearest integer. Compute the four fundamental subspaces associated with the matrix Y and illustrate their significance. Provide a **complete and neat diagram** showing the orthogonal complement structure and interactions among the four subspaces. (3 marks)

Q3. Explain the gradient descent algorithm in detail. Mention:

- (i) Effect of low learning rate on the convergence of gradient descent.
- (ii) Effect of high learning rate on the convergence of gradient descent.
- (iii) How the concept of vectorization is used in the implementation of gradient descent.

Explain the terminologies: bias, weights, loss function (error or cost function), epoch, learning rate, activation function, underfitting, overfitting, regularisation, regularisation rate.

(2 marks)

Q4. List all the matrix decompositions that you have studied. For each decomposition, explain its significance; key features; and limitations. Mention when each of these decompositions are to be used.

(2 marks)

Q5. Which lab session did you find most impactful? Elaborate on the reasons behind your choice, highlighting the important features that made the session stand out.

(1 Mark)